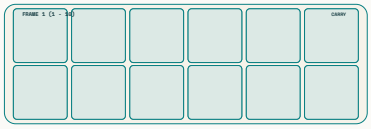
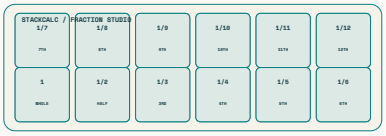
K–12 TEACHER CURRICULUM & LEARNING LAB GUIDE

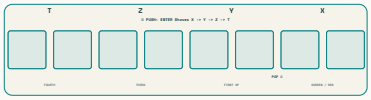
Book-Aligned Lesson Plans, 3D-Printable Gridfinity Manipulative Setups, RPN Stack Execution Traces & Reproducible Student Lab Sheets (Pre-K to Grade 12 / AP)

ALIGNED TO OUR FLAGSHIP ILLUSTRATED BOOKS & MAJOR TEXTBOOK SERIES

- **Elementary & Middle School Companion: From Blocks to the Stack (SC-6 Elementary Calculator)**
- **High School, AP & Engineering Companion: Thinking in the Stack (SC-32 Scientific Calculator)**
- Direct Pacing Concordances: Singapore Dimensions Math (PK–8), Bridges in Mathematics (K–5), Big Ideas Math (Grades 6–12), Common Core (CCSS), TEKS, and AP Calculus / Statistics / CS







WHAT IS INSIDE THIS SPIRAL-BOUND TEACHER EDITION

1. 12 Complete 3-Page Curricular Units (Units 01–12): Core Lesson, Extension Lab & Student Sheet.
2. Historical & Real-World Book Stories: Every unit opens with its story from our flagship books.
3. Exact Hardware Trapezoid Keycaps: Every keystroke sequence & 4-register (T,Z,Y,X) trace verified.
4. Complete Worked Answer Keys & Rubrics: Solutions for all 12 Core, 12 Extension & 48 Lab Problems.
5. Extended Back Matter Appendices: 3D Slicer Profiles, Manipulative BOM, Keystroke Reference & Log.
6. **100% Lay-Flat Coil Margin: Engineered with a 0.50 in (12.7 mm) coil-punch margin for desk use.**

STACKCALC LEARNING SYSTEMS • OPEN-SOURCE HARDWARE & MATHEMATICS EDUCATION

Web Portal & Printable Chapter PDFs: https://stackcalc.io/curriculum/teacher-guide/

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PUBLICATION METADATA & LULU COIL-BOUND PRINT SPECIFICATIONS

Title: StackCalc K–12 Teacher Curriculum & Learning Lab Guide (Lay-Flat Spiral Edition)

Collection 01 • Revision 0.3.0 • SKU: STK-CURR-GUIDE-SPIRAL • Trim: US Letter (8.50 × 11.00 in / 215.9 × 279.4 mm)

Binding: Heavy-Duty Lay-Flat Plastic Coil / Wire-O via Lulu Print-on-Demand (360° Desk Fold Verified)

Margin Geometry: >= 0.55 in (14.0 mm) Binding-Edge Margin (Exceeds Lulu 0.50 in / 12.7 mm Coil Punch Gate)

Paper Stock: 60#–80# Uncoated Archival White Paper (Engineered for Classroom Pencil, Eraser & Dry-Erase Sleeve Use)

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Individual 4-Page Unit PDFs for Parents & Teachers: https://stackcalc.io/curriculum/teacher-guide/

OFFICIAL ECOSYSTEM CHANNEL DIRECTORY (WHERE TO GET BOOKS vs. MANIPULATIVE BLOCKS)

1. SPIRAL-BOUND CURRICULUM BOOKS & WORKBOOKS — LULU PRINT-ON-DEMAND

- Order lay-flat coil-bound editions of this K–12 Teacher Curriculum Guide, the 108-page Big Ideas Math Middle School Student Workbook, and the Proficiency Trials Book on Lulu: https://www.lulu.com/spotlight/stackcalc

2. FLAGSHIP ILLUSTRATED TRADE BOOKS (PAPERBACK & HARDCOVER) — AMAZON

- Order From Blocks to the Stack (SC-6 Elementary) & Thinking in the Stack (SC-32 Scientific) in 6×9 trade paperback and hardcover editions on Amazon: https://link.amazon/B0f5cP0Ur

3. HANDCRAFTED 3D-PRINTED MANIPULATIVE BOXED KITS & HARDWARE — ETSY STOREFRONT

- Physical 3D-printed Gridfinity manipulative kits & calculator hardware kits (no books sold on Etsy): https://stackcalcmath.etsy.com

4. FACTORY-PRINTED MANIPULATIVES (SLANT 3D) & 100% FREE OPEN-SOURCE STLS (PRINTABLES)

- Slant 3D On-Demand Kits: https://teleportpod.com/portals/profile/StackCalc | Free STLs: https://www.printables.com/@StackCalc_5593834

School District & Institutional Purchase Orders (Net-30)

Schools and districts can order classroom manipulative fleets and bulk Lulu spiral workbooks via formal Purchase Order (orders@stackcalc.io). Download our W-9 & PO Packet at https://stackcalc.io/procurement/.

FRONT MATTER • MASTER TABLE OF CONTENTS

Table of Contents & Book Story Directory

Complete page index for Front Matter, Pacing Concordances, Units 01–12, Answer Keys, and Back Matter Appendices.

Section / Unit	Grades & Kit	Aligned Flagship Book Story & Chapter	Pages
Front Matter (Title, Colophon, TOC	All Grades	Lulu Spiral Specs, Reproduction License & CRA+RP	**pp. i–iv**
Hardware Overview & 3 Pacing Conco	PK–12 / AP	Singapore Dimensions, Bridges & Big Ideas Math I	**pp. 1–4**
Unit 01: Counting, Cardinality & T	Pre-K – • BASE-10	How Ten Fingers & an Empty Shell Created P (From	**pp. 5–7**
Unit 02: Part-Part-Whole & Doubles	Grades K • DOMI-01	Spotting Number Partners at a Glance: Subi (From	**pp. 8–10**
Unit 03: Ten-Frames: Complements t	Grade 1 • BASE-10	Making a Ten First (8 + 5 = 10 + 3 = 13) & (From	**pp. 11–13**
Unit 04: Number Racks (Rekenrek):	Grades 1 • RACK-01	Two Hands, One Workbench: Why 5+5 Bead Row (From	**pp. 14–16**
Unit 05: The 4-Register Stack: Fou	Grades 3 • PLACE-0	Inside the 4-Floor Number Elevator (T, Z, (From	**pp. 17–19**
Unit 06: Fraction Studio: Height E	Grades 4 • FRAC-01	The Pharaoh's Bakery, Pythagoras's Guitar (From	**pp. 20–22**
Unit 07: Postfix Evaluation & Chai	Grades 5 • COUNT-0	Why 24 Is the Champion Packing Number, The (From	**pp. 23–25**
Unit 08: Tactile Algebra: Area Mod	Grades 7 • ALGB-01	From Al-Khwarizmi's Physical Squares to th (From	**pp. 26–28**
Unit 09: Trigonometry, Vectors & P	Grades 9 • VOLV-01	A Stick in Alexandria: How Eratosthenes Me (Thin	**pp. 29–31**
Unit 10: Exponential Growth, Compo	Grades 1 • PRIME-0	The Scottish Baron Who Doubled the Lifespa (Thin	**pp. 32–34**
Unit 11: Combinatorics, Probabilit	Grades 1 • PROB-01	Sir Francis Galton's 1873 Bean Machine, Th (From	**pp. 35–37**
Unit 12: Digital Logic, Base Conve	Grades 9 • BASE-10	Leibniz's Binary Medallion, Claude Shannon (Thin	**pp. 38–40**
Master Solutions & Keystrokes (Par	Units 01–12	Complete Core, Extension & 48 Student Lab Sheet	**pp. 41–42**
Appendix A: 3D Printing & Slicer P	Makerspace	PLA/PETG Settings, 0.6mm Debossing & Fit Toleran	**p. 43**
Appendix B: Manipulative Bill of M	All 12 Kits	Millimeter Dimensions, Piece Counts & Math Invar	**p. 44**
Appendix C: SC-6 & SC-32 Keystroke	SC-6 / SC-32	4-Register Stack (T,Z,Y,X), Two-Dot Fractions &	**p. 45**
Appendix D: Classroom Pacing & Sta	Teacher Log	Photocopiable 12-Unit Pacing Tracker & Notes Gri	**p. 46**

FRONT MATTER • CLASSROOM METHODOLOGY & PACING

How to Teach with the Spiral Edition

The 4-step Concrete-Representational-Abstract + RPN verification protocol and turnkey pacing schedules.

1. THE 4-STEP HANDS-ON BLOCK TO STACKCALC VERIFICATION LOOP

STEP 1: STORY & CONCRETE BLOCKS (10 MIN)

Open with the unit's Flagship Book Story. Students build the problem physically on their 42 mm Gridfinity stage (Ten-Frames, 84mm Fraction Towers, Algebra Tiles, Volvelle) before touching a pencil or calculator.

STEP 2: PICTORIAL SKETCH & PREDICTION (10 MIN)

Students sketch their block model in Box 1 of the Student Lab Worksheet, write their paper-and-pencil strategy in Box 2, and write their predicted answer on the Prediction line.

STEP 3: STACKCALC RPN VERIFICATION (10 MIN)

Students key their operands and operators into the SC-6 Elementary or SC-32 Scientific calculator, observing how each step transforms the 4-level stack (T, Z, Y, X) and checking [x] Checked on StackCalc.

STEP 4: EXTENSION & ERROR DIAGNOSIS (15 MIN)

When a student's calculator display differs from their written prediction, they inspect Register Y and Register X alongside their physical blocks to self-diagnose regrouping, denominator, or sign slips.

2. FOUR FLEXIBLE CLASSROOM & HOMESCHOOL PACING SCHEDULES

Schedule A: Weekly Friday STEM Lab (36 Weeks)

Teach 1 Unit every 3 weeks: Week 1 = Core Lesson (Page A), Week 2 = Extension Investigation (Page B), Week 3 = Student Lab Worksheet (Page C) + Timed Proficiency Sprint.

Schedule B: 3-Day Textbook Supplement Block

When your primary textbook (Singapore Dimensions, Bridges, or Big Ideas Math) reaches a key topic, run Page A on Day 1, Page B on Day 2, and Page C as an independent station on Day 3.

Schedule C: Math Center / Station Rotation (20 Min)

Fold this spiral guide 360° flat at the manipulative table as a self-guided station prompt while small groups rotate through physical blocks, sketch workspaces, and SC-6/SC-32 check.

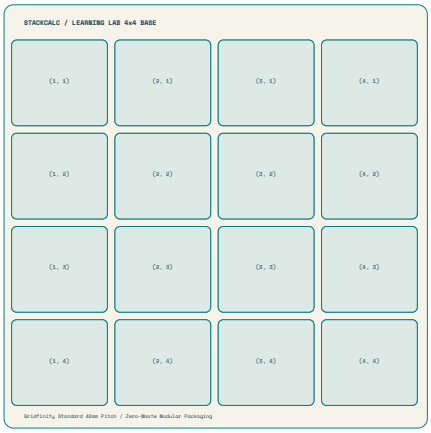
Schedule D: Homeschool & Family Co-Op Loop

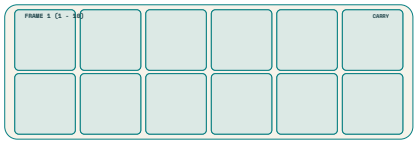
Read the aligned chapter story in From Blocks to the Stack or Thinking in the Stack together, build the physical block model on the table, and complete the 4-problem Student Lab Sheet.

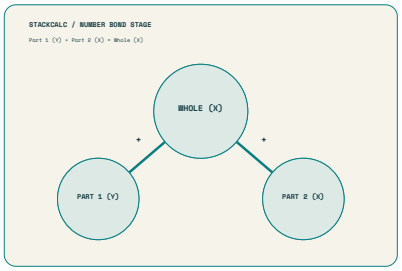
MODULAR HARDWARE & CURRICULUM GUIDE

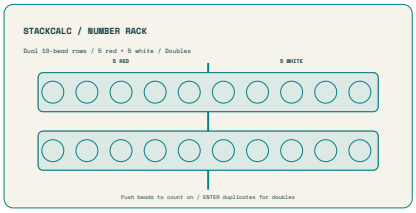
Make the Math Tangible.

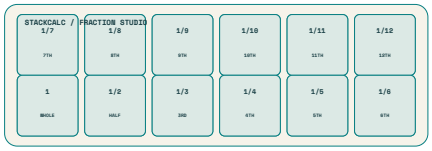
Modular Gridfinity manipulatives and chapter-by-chapter lesson plans from Pre-K to Grade 12+.

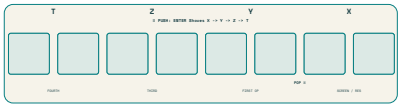












ZERO-WASTE MODULAR PACKAGING: THE BOX IS THE LAB

All activity stages adhere to the open-source 42.0 mm Gridfinity standard. The storage baseplate provides rock-solid physical support during classroom use and nests all manipulatives for compact storage. No disposable packaging.

SINGLE-OPERATION FDM PRINTING: 0.6 MM DEBOSSED TEXT

Numerals, labels, and coordinate grids are debossed directly into top surfaces (0.6 mm deep, exactly 3 layers at 0.20 mm). Prints cleanly in PLA or PETG without secondary laser marking or multi-material setup.

FULL EDUCATIONAL SPAN: PRE-K TO AP COMPUTER SCIENCE & CALCULUS

StackCalc connects early arithmetic (counting on, number bonds, ten-frames, rekenreks) directly to middle school algebra, high school trigonometry, exponential growth, combinatorics, and computer architecture.

TEXTBOOK PACING & LESSON INDEX

Singapore Dimensions Math Master Pacing Concor

Find your current Dimensions Math PK–8 book, chapter, and lesson number to locate the matching Gridfinity stage and book pages.

Grade	Textbook Ref	Topic & Objective	Manipulative	Keystrokes	Book Pages
Pre-K	Dimensions PK Book A,	Counting to 5 with Concrete	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Pre-K	Dimensions PK Book B,	Comparing Quantities: More o	Ten-Frame Stage	[5] [ENTER] [2]	pp. 5–7
Kindergarten	Dimensions KA, Ch 1, L	One-to-One Counting & Frames	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Kindergarten	Dimensions KA, Ch 2, L	Zero and Empty Sets	Ten-Frame Stage	[0] / [C]	pp. 5–7
Kindergarten	Dimensions KA, Ch 4, L	Number Bonds of 5	Math Domino Stu	[3] [ENTER] [2]	pp. 8–10
Kindergarten	Dimensions KB, Ch 7, L	Addition within 10 on the St	Math Domino Stu	[4] [ENTER] [3]	pp. 8–10
Kindergarten	Dimensions KB, Ch 8, L	Subtraction as Missing Part	Math Domino Stu	[1][0] [ENTER] [	pp. 8–10
Grade 1	Dimensions 1A, Ch 1, L	Counting & Comparing Numbers	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Grade 1	Dimensions 1A, Ch 2, L	Making Number Bonds to 6 and	Math Domino Stu	[5] [ENTER] [3]	pp. 8–10
Grade 1	Dimensions 1A, Ch 2, L	Making Number Bonds to 10	Math Domino Stu	[7] [ENTER] [3]	pp. 8–10
Grade 1	Dimensions 1A, Ch 2, L	Decomposing Numbers & Missin	Math Domino Stu	[1][0] [ENTER] [	pp. 8–10
Grade 1	Dimensions 1A, Ch 3, L	Addition as Part-Part-Whole	Math Domino Stu	[PART 1] [ENTER]	pp. 8–10
Grade 1	Dimensions 1A, Ch 4, L	Subtraction as Comparing Set	Ten-Frame Stage	[LARGE] [ENTER]	pp. 5–7
Grade 1	Dimensions 1A, Ch 7, L	Ten-Frames & Place Value (11	Ten-Frame Stage	[1][0] [ENTER] [	pp. 11–13
Grade 1	Dimensions 1A, Ch 8, L	Making 10 to Add	Ten-Frame Stage	[1][0] [ENTER] [	pp. 11–13
Grade 1	Dimensions 1A, Ch 8, L	Bridging Through 10	Ten-Frame Stage	[8] [ENTER] [2]	pp. 11–13
Grade 1	Dimensions 1B, Ch 12,	Multiplication as Equal Grou	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 1	Dimensions 1B, Ch 14,	Halves & Fourths (Equal Shar	Fraction Studio	[. 1 . 2] / [. 1	pp. 20–22
Grade 2	Dimensions 2A, Ch 2, L	Mental Addition with Number	Math Domino Stu	[P1] [ENTER] [P2	pp. 8–10
Grade 2	Dimensions 2A, Ch 4, L	Doubles & Multiplying by 2	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 2	Dimensions 2B, Ch 6, L	Near-Doubles Strategies (Dou	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 2	Dimensions 2B, Ch 7, L	Unit Fractions (Halves, Thir	Fraction Studio	Compare 42mm / 2	pp. 20–22
Grade 3	Dimensions 3A, Ch 2, L	Multi-Digit Addition on Stac	Place Value Stu	[A] [ENTER] [B]	pp. 17–19

Using the Singapore Math Pacing Index

Locate your lesson above for this week's class. Turn to the listed pages in this guide: Page 17 provides the Core Lesson Plan, Page 18 the Extension, and Page 19 the reproducible Student Activity Sheet.

TEXTBOOK PACING & LESSON INDEX

Bridges in Mathematics Master Pacing Concordan

Find your current unit, module, and daily session number to locate the matching Gridfinity stage and book pages.

Grade	Textbook Ref	Topic & Objective	Manipulative	Keystrokes	Book Pages
Kindergarten	Unit 1, Mod 1, Session	Welcome to Math: Counting Ob	Ten-Frame Stage	1 peg = 1 count	pp. 5–7
Kindergarten	Unit 1, Mod 1, Session	Counting & Five-Frames	Ten-Frame Stage	Top row = 5	pp. 5–7
Kindergarten	Unit 1, Mod 2, Session	Ten-Frames & Counting On	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Kindergarten	Unit 1, Mod 2, Session	Zero & The Empty Ten-Frame	Ten-Frame Stage	[0] / [C]	pp. 5–7
Kindergarten	Unit 1, Mod 3, Session	The Step Counter: Adding One	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Kindergarten	Unit 2, Mod 1, Session	Comparing Sets: Which Has Mo	Ten-Frame Stage	[5] [ENTER] [2]	pp. 5–7
Kindergarten	Unit 2, Mod 2, Session	Number Bonds to 5	Math Domino Stu	Part-Part-Whole	pp. 8–10
Kindergarten	Unit 4, Mod 2, Session	Part-Part-Whole Addition	Math Domino Stu	[P1] [ENTER] [P2	pp. 8–10
Grade 1	Unit 1, Mod 2, Session	Ten-Frames & Subitizing	Ten-Frame Stage	Instant recognit	pp. 11–13
Grade 1	Unit 2, Mod 1, Session	Combinations to Ten	Ten-Frame Stage	[1][0] [ENTER] [	pp. 11–13
Grade 1	Unit 2, Mod 1, Session	Finding the Missing Addend	Math Domino Stu	[WHOLE] [ENTER]	pp. 8–10
Grade 1	Unit 2, Mod 3, Session	Making 10 Addition Strategie	Ten-Frame Stage	Bridging 8 + 5 =	pp. 11–13
Grade 1	Unit 3, Mod 1, Session	Introducing the Number Rack	Number Rack	5 red + 5 white	pp. 14–16
Grade 1	Unit 3, Mod 1, Session	Double Flips on the Number R	Number Rack	[6] [ENTER] [+]	pp. 14–16
Grade 1	Unit 3, Mod 1, Session	Doubles Plus or Minus One	Number Rack	[6] [ENTER] [+]	pp. 14–16
Grade 1	Unit 3, Mod 2, Session	5-Structure on the Rack	Number Rack	Subitizing 5 gro	pp. 14–16
Grade 1	Unit 6, Mod 1, Session	Two-Row Rack Comparisons	Number Rack	Difference betwe	pp. 14–16
Grade 1	Unit 7, Mod 1, Session	Place Value: Tens & Ones	Ten-Frame Stage	10 + N on the st	pp. 11–13
Grade 2	Unit 1, Mod 1, Session	Number Rack Fact Fluency	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 2	Unit 1, Mod 1, Session	Near-Doubles Fact Strategies	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 2	Unit 1, Mod 2, Session	Combinations to 20	Number Rack	Dual row combina	pp. 14–16
Grade 2	Unit 3, Mod 2, Session	Two-Digit Subtraction on Sta	SC-32 Hardware	[Y] [ENTER] [X]	pp. 17–19
Grade 3	Unit 1, Mod 2, Session	Multi-Step Addition & Subtra	Place Value Stu	Chained operatio	pp. 17–19

Using the Bridges in Mathematics Pacing Index

Locate your lesson above for this week's class. Turn to the listed pages in this guide: Page 17 provides the Core Lesson Plan, Page 18 the Extension, and Page 19 the reproducible Student Activity Sheet.

TEXTBOOK PACING & LESSON INDEX

Big Ideas Math & AP Master Pacing Concordance

Find your current grade, chapter, and section number to locate the matching Gridfinity stage and book pages.

Grade	Textbook Ref	Topic & Objective	Manipulative	Keystrokes	Book Pages
Kindergarten	Chapter 1, Section 1.2	Count and Write 1 to 5	Ten-Frame Stage	[1] [ENTER] [1]	pp. 5–7
Kindergarten	Chapter 3, Section 3.1	Model and Count 6 to 10	Ten-Frame Stage	5 top + N bottom	pp. 5–7
Kindergarten	Chapter 5, Section 5.1	Number Bonds to 10 on the Fr	Math Domino Stu	Part 1 in Y, Par	pp. 8–10
Kindergarten	Chapter 7, Section 7.2	Represent Addition as Puttin	Math Domino Stu	[P1] [ENTER] [P2	pp. 8–10
Grade 1	Chapter 1, Section 1.4	Part-Part-Whole Addition	Math Domino Stu	[4] [ENTER] [3]	pp. 8–10
Grade 1	Chapter 2, Section 2.3	Think Addition to Subtract	Math Domino Stu	[1][0] [ENTER] [	pp. 8–10
Grade 1	Chapter 3, Section 3.1	Doubles Strategies with ENTE	Number Rack	[6] [ENTER] [+]	pp. 14–16
Grade 1	Chapter 3, Section 3.2	Doubles Plus 1 & Doubles Min	Number Rack	[6] [ENTER] [+]	pp. 14–16
Grade 1	Chapter 3, Section 3.4	Make a 10 to Add on Ten-Fram	Ten-Frame Stage	[1][0] [ENTER] [	pp. 11–13
Grade 1	Chapter 4, Section 4.2	Use 10 to Subtract	Ten-Frame Stage	Bridging through	pp. 11–13
Grade 2	Chapter 1, Section 1.2	Mental Math: Doubles & Near-	Number Rack	[N] [ENTER] [+]	pp. 14–16
Grade 2	Chapter 1, Section 1.4	Make a Ten Addition Strategi	Ten-Frame Stage	Complements on f	pp. 11–13
Grade 2	Chapter 3, Section 3.3	Two-Digit Addition with Lift	Place Value Stu	Register lift on	pp. 17–19
Grade 3	Chapter 1, Section 1.3	Multiplication as Repeated A	Place Value Stu	[N] [ENTER] [+]	pp. 17–19
Grade 3	Chapter 8, Section 8.2	Equal Parts & Unit Fractions	Fraction Studio	Block heights co	pp. 20–22
Grade 3	Chapter 8, Section 8.5	Equivalent Fractions on Bloc	Fraction Studio	Equal tower heig	pp. 20–22
Grade 4	Chapter 3, Section 3.4	Solve Multi-Step Word Proble	SC-32 Hardware	[15] [ENTER] [6]	pp. 17–19
Grade 4	Chapter 3, Section 3.5	Order of Operations Without	SC-32 Hardware	Auto-lift retain	pp. 17–19
Grade 4	Chapter 8, Section 8.3	Add & Subtract Common Fracti	Fraction Studio	[. 1 . 4] [ENTER	pp. 20–22
Grade 4	Chapter 8, Section 8.6	Mixed Numbers: Two-Dot Entry	Fraction Studio	[2 . 1 . 2] = 2	pp. 20–22
Grade 5	Chapter 5, Section 5.2	Add Fractions with Unlike De	Fraction Studio	Common tower hei	pp. 20–22
Grade 5	Chapter 6, Section 6.2	Multiply Fractions & Height	Fraction Studio	[. 3 . 4] [ENTER	pp. 20–22
Grade 5	Chapter 6, Section 6.5	Divide Fractions with Recipr	Fraction Studio	[¹/▯] inverts di	pp. 20–22

Using the Big Ideas Math Pacing Index

Locate your lesson above for this week's class. Turn to the listed pages in this guide: Page 20 provides the Core Lesson Plan, Page 21 the Extension, and Page 22 the reproducible Student Activity Sheet.

UNIT 1 CORE LESSON / BRIDGES IN MATHEMATICS (PRE-K – KINDERGARTEN)

1-to-1 Correspondence: The Concrete Step Counte

Numbers to 5 & 10 • Standard: CCSS.MATH.K.CC.B.4

BOOK STORY (BASE-10): How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty tray had its own symbol. Dropping 1 counter at a time into the Ten-Frame while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Counting, Subitizing & Place Value + Appendi

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Ten-Frame Stage (4x2 Gridfinity)

Place 1 peg into the first well of the Ten-Frame. Have students observe the single object.

2. STUDENT PREDICTION (BEFORE PRESSING)

When we place 1 more peg into the next well, how many pegs will be on our frame? (Prediction: 2)

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow:



Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
1	1	First peg placed: 1 enters register X
ENTER	1	1 copied to register Y: holds the current count
1	1	Second peg placed: 1 enters register X
+	2	Press [+]: 1 + 1 = 2! Screen verifies 2 pegs on board
1	1	Third peg placed in frame: 1 enters X
+	3	Press [+]: 2 + 1 = 3! Frame and screen match perfectly

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Each physical counter corresponds to a single increment. The calculator's stack preserves the running count in Y while new items enter X.

Classroom Investigation Problem:

Place 4 pegs on your Ten-Frame. Add 1 more peg. Write the keystrokes to show 4 + 1 on the calcu

UNIT 1 EXTENSION / SINGAPORE MATH (PRE-K – KINDERGARTEN)

More, Less & Zero: Finding Differences

Comparing Sets & Magnitude • Standard: CCSS.MATH.K.CC.C.6

1. ADVANCED INVESTIGATION SETUP

Manipulative: Ten-Frame Stage (4x2 Gridfinity)

Place 5 pegs in the top row (Row 1) and 2 pegs in the bottom row (Row 2).

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Look at the two rows. Which row has more? How many more pegs are in Row 1 than Row 2?

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
	**5**	Row 1 quantity (5) enters register X
	**5**	5 lifted to register Y (first set)
	**2**	Row 2 quantity (2) enters register X
	**3**	Minus subtracts X from Y: Row 1 has 3 more pegs!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Comparison subtraction answers 'how many more?'. Register Y holds the larger group and register X holds the smaller group.

Extension Investigation Challenge:

Row 1 has 6 pegs and Row 2 has 2 pegs. How many more pegs in Row 1? Write the keystrokes.

STUDENT LAB WORKSHEET • UNIT 1

Counting, Cardinality & The One-More Step Coun

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW COUNTING MAPS TO THE CALCULATOR

- Every peg placed on the frame matches one count on the calculator screen.
- An empty ten-frame represents zero [0]. Press [C] to start a fresh count.
- To step count by 1: press [1] [ENTER] [1] [+], then press [1] [+] to see the count advance!

Calculator Shortcut: [1] [ENTER] [1] [+] = 2, then [1] [+] = 3, [1] [+] = 4...

Problem 1: Place 3 pegs on your frame. Add 2 more. Write the keystrokes to find the total.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: A frame has 5 pegs. If you remove 2 pegs, how many remain?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Clear the calculator by pressing [C]. What number represents an empty ten-frame?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 2 CORE LESSON / SINGAPORE MATH (GRADES K-1)

Composing Numbers: Part + Part = Whole

Number Bonds & Part-Part-Whole • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY (DOMI-01): Spotting Number Partners at a Glance: Subitizing Domino Pips & Friends of Ten

Your brain has a built-in superpower called subitizing (from the Latin 'subitus', meaning 'sudden'): you can recognize 1 to 5 domino pips instantly without counting one by one! In From Blocks to the Stack, the left half of a domino sits on Shelf Y (Part 1), the right half sits on Shelf X (Part 2), and pressing [+] snaps both halves together into the Whole.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Number Bonds, Ch. 8: Missing Addend Mysterie

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Math Domino Studio (12-Frame Tray)

Place the [4|3] domino in the 12-frame tray. Count 4 pips on the left (Part 1) and 3 pips on the right (Part 2).

2. STUDENT PREDICTION (BEFORE PRESSING)

Combine both parts on the tray: 4 + 3 = 7 total pips.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**4**	Part 1 (4) enters register X
	**4**	4 copied up to register Y; represents Part 1
	**3**	Part 2 (3) enters register X
	**7**	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack has dedicated registers for parts: register Y holds Part 1, register X holds Part 2, and [+] merges them into the Whole.

Classroom Investigation Problem:

Pick the [5|4] domino tile. Place it on your 12-frame tray to find the whole. Write the keystro

Decomposing the Whole: Finding the Missing Par

Developing Number Sense • Standard: CCSS.MATH.1.OA.B.4

1. ADVANCED INVESTIGATION SETUP

Manipulative: Math Domino Studio (12-Frame Tray)

Place 10 counters in the 12-frame tray. Cover or slide 6 counters to represent known Part 1.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

How many counters remain uncovered in the tray? (10 - 6 = 4).

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
	**1**	Enter Whole (10)
	**10**	10 in register X
	**10**	Whole (10) lifted to register Y
	**6**	Known Part 1 (6) enters register X
	**4**	Minus subtracts known Part from Whole: Missing Part is 4!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Finding a missing addend is inverse addition. Placing the Whole in Y and subtracting known Part in X yields the missing part in X.

Extension Investigation Challenge:

The whole is 8. Part 1 is 5. Find the missing Part 2 using the calculator.

STUDENT LAB WORKSHEET • UNIT 2

Part-Part-Whole & Doubles: The Math Domino Stu

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW DOMINOES FIT THE STACK

- Part 1 lives in register Y. Part 2 enters register X.
- Pressing [+] combines both parts into the Whole in register X.
- To find a missing part: start with Whole in Y, enter Part 1 in X, and press [-]!

Calculator Shortcut: [PART 1] [ENTER] [PART 2] [+] = [WHOLE]

Problem 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Part 1 is 7, Part 2 is 2. What is the Whole?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: The Whole is 10. Part 1 is 7. What is Part 2?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 3 CORE LESSON / BIG IDEAS MATH (GRADE 1)

Complements to 10: Making Ten

Addition Strategies to 10 • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY (BASE-10): Making a Ten First (8 + 5 = 10 + 3 = 13) & The Double-Height '10' Carry Block

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students slide 2 dots from the right tray into the 2 empty pockets of the left tray to complete a full 10, then physically swap all 10 loose dots for one double-height '10' Carry Block.
Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Making a Ten First & Trading 10 Ones + Appen

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Ten-Frame Bin (3x1 Gridfinity)

Place 7 unnumbered counters into the Ten-Frame: fill the top row of 5, then place 2 in the bottom row.

2. STUDENT PREDICTION (BEFORE PRESSING)

Look at the empty wells without counting one-by-one. 3 empty spaces mean 7 + 3 = 10.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**1**	Enter 1
	**10**	Full frame capacity (10) in X
	**10**	10 lifted to Y
	**7**	Occupied counters (7) in X
	**3**	10 - 7 = 3: Exactly 3 empty wells to make 10!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The ten-frame physically structures 10 as two groups of 5. Subtraction verifies the visual complement.

Classroom Investigation Problem:

A ten-frame has 6 counters. How many more counters are needed to make 10?

UNIT 3 EXTENSION / SINGAPORE MATH (GRADE 1)

Bridging Through 10: The Make-Ten & Carry Stra

Addition Within 20 • Standard: CCSS.MATH.1.OA.C.6

1. ADVANCED INVESTIGATION SETUP

Manipulative: Ten-Frame Bin (3x1 Gridfinity)

Place 8 unnumbered counters in Ten-Frame 1. Prepare 5 counters beside the board.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Take 2 from the 5 to fill Ten-Frame 1 to 10. Trade the full frame for one '10' block in the Carry Bay! 3 loose counters remain. 10 + 3 = 13.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
	**8**	8 counters on frame
	**8**	8 lifted to Y
	**2**	2 counters borrowed to make 10
	**10**	8 + 2 = 10 (Full frame traded for '10' carry block!)
	**3**	Remaining 3 counters
	**13**	10 + 3 = 13 total!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Bridging decomposes 8 + 5 into (8 + 2) + 3. On RPN, intermediate sums remain in the accumulator register.

Extension Investigation Challenge:

Use the Make-Ten strategy to solve 9 + 6 on your calculator.

STUDENT LAB WORKSHEET • UNIT 3

Ten-Frames: Complements to 10 & 5-Structure

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TEN-FRAME 5-STRUCTURE & CARRY RULES

- The 2x5 grid holds unnumbered counter blocks. Spatial position builds visual subitizing without numeral distraction.
- Subitizing: 7 is seen immediately as 5 on top plus 2 on bottom.
- Dual 10-Frame Trays: Students use two dedicated 10-frame trays (strictly 10 slots each, solid bottoms) side-by-side on the c

Calculator Shortcut: 10 [ENTER] [N] [-] = [COMPLEMENT TO 10]

Problem 1: What is the complement to 10 for 4? Write the keystrokes.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: A ten-frame has 9 counters. How many to make 10?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Solve 8 + 6 using Make-Ten on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 4 CORE LESSON / BRIDGES IN MATHEMATICS (GRADES 1–2)

Doubles Strategy: The ENTER Duplicate Key

Number Racks & Fact Strategies • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY (RACK-01): Two Hands, One Workbench: Why 5+5 Bead Rows Mirror the Stack & Instant [ENTER]

Designed in the Netherlands by math educator Adrian Treffers, the Rekenrek ('calculating rack') splits each row into 5 red and 5 white beads—mirroring the 5 fingers on your left and right hands. In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Shelf X onto Shelf Y just like matching the bottom bead row to the top row for instant doubles!
Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 1: Meet Your Math Workbench & Ch. 2: What Happe

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Number Rack Stage (4x2 Gridfinity)

Push 6 beads on the top row (5 red, 1 white). Push 6 beads on the bottom row.

2. STUDENT PREDICTION (BEFORE PRESSING)

Observe the matching rows. 6 + 6 = (5 + 5) + (1 + 1) = 10 + 2 = 12.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**6**	Top row quantity (6) enters register X
	**6**	ENTER duplicates 6 into register Y! Both rows now on stack
	**12**	Plus combines top and bottom rows: 6 + 6 = 12!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

ENTER creates an instant identical copy of register X in register Y. This perfectly represents a physical double.

Classroom Investigation Problem:

Push 7 beads on each row of the Number Rack. Use [ENTER] [+] to double it on the calculator.

Near-Doubles: Doubles Plus One

Fact Fluency & Near Doubles • Standard: CCSS.MATH.2.OA.B.2

1. ADVANCED INVESTIGATION SETUP

Manipulative: Number Rack Stage (4x2 Gridfinity)

Push 7 beads on top row (5 red, 2 white). Push 8 beads on bottom row (5 red, 3 white).

2. HYPOTHESIS & MATHEMATICAL PREDICTION

7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
	**7**	Enter base number 7
	**7**	Duplicate 7
	**14**	Double: 7 + 7 = 14
	**1**	Near-double difference (1)
	**15**	14 + 1 = 15!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Near-doubles build upon doubles fluency. The calculator executes the base double, then accumulates the offset.

Extension Investigation Challenge:

Use the near-doubles strategy to solve 6 + 7 on the calculator.

STUDENT LAB WORKSHEET • UNIT 4

Number Racks (Rekenrek): 5-Structure & Doubles

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: NUMBER RACK & ENTER DUPLICATE RULES

- Each row has 5 red beads and 5 white beads.
- Pushing beads to the left counts them. 5 red + 5 white = 10 per row.
- Doubles shortcut: Enter a number, press [ENTER] to duplicate, then press [+]!

Calculator Shortcut: [N] [ENTER] [+] = [DOUBLES (2 * N)]

Problem 1: Calculate double 8 using [ENTER] [+].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 2: Solve near-double 8 + 9: double 8 then add 1.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 3: Calculate double 9 using [ENTER] [+].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 4: Solve 5 + 6 using doubles plus one.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

UNIT 5 CORE LESSON / SINGAPORE MATH (GRADES 3–4)

Multi-Step Word Problems Without Parentheses

Multi-Step Word Problems • Standard: CCSS.MATH.4.OA.A.3

BOOK STORY (PLACE-01): Inside the 4-Floor Number Elevator (T, Z, Y, X) & Young Gauss's 10-Second Shor

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—folding the list into 50 pairs of 101! In Chapters 2 & 4 of From Blocks to the Stack, students discover how the 4-Floor Number Elevator (Shelves X, Y, Z, T) automatically lifts finished answers up to Shelf Y when you type the next number, solving multi-step stories with zero parentheses.
Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 2: The Number Elevator & Ch. 4: Young Gauss's 1

1. CONCRETE MANIPULATIVE SETUP

Manipulative: SC-32 Hardware LCD Stack & Place Value Studio

Use Place Value Studio unit blocks to represent quantities alongside the SC-32 4-line LCD screen.

2. STUDENT PREDICTION (BEFORE PRESSING)

Problem: (8 - 3) * 2. Predict: 8 - 3 = 5, then 5 * 2 = 10.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**8**	8 in X
	**8**	8 copied to Y
	**3**	3 entered into X
	**5**	8 - 3 = 5 in X
	**2**	Automatic lift! 5 lifts to Y; 2 enters X
	**10**	5 * 2 = 10 in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Unlike traditional algebraic calculators that require parenthesis keys, RPN evaluates inner expressions first, lifting results naturally.

Classroom Investigation Problem:

A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes f

UNIT 5 EXTENSION / BIG IDEAS MATH (GRADES 3-4)

Automatic Stack Lift & Register Retention

Order of Operations & Stack Lift • Standard: CCSS.MATH.4.OA.A.3

1. ADVANCED INVESTIGATION SETUP

Manipulative: SC-32 Hardware LCD Stack & Place Value Studio

Observe the 4-line hardware LCD screen as values automatically lift into register Y: (12 / 4) + 7.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
12	**12**	12 entered in X
ENTER	**12**	12 copied to Y
4	**4**	4 enters X
/	**3**	12 / 4 = 3 in X
7	**7**	7 entered: 3 lifts to Y automatically!
+	**10**	3 + 7 = 10!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Automatic lift prevents unnecessary ENTER presses between sequential chained operations.

Extension Investigation Challenge:

Calculate (24 / 6) + 15 on the calculator without pressing ENTER twice.

STUDENT LAB WORKSHEET • UNIT 5

The 4-Register Stack: Four Operations & Automa

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, *, /) combine Y and X, dropping Z to Y and T to Z.

Calculator Shortcut: [Y] [ENTER] [X] [OP] -> Drops stack and displays result in X

Problem 1: Calculate (14 - 5) * 3 on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Calculate 30 / (2 + 3): add denominator first!

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Calculate (9 + 7) / (8 - 4) using 4 registers.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 6 CORE LESSON / SINGAPORE MATH (GRADES 4–5)

Mixed Numbers & Height Equivalence: 3/4 + 2 1/

Fraction Addition & Mixed Numbers • Standard: CCSS.MATH.5.NF.A.1

BOOK STORY (FRAC-01): The Pharaoh's Bakery, Pythagoras's Guitar Strings & Why Your Fraction Towers S

Four thousand years ago, Egyptian scribes had to divide 3 loaves of bread fairly among 4 workers—and instead of cutting every loaf into quarters, they gave each worker 1/2 + 1/4! In Chapter 5 of From Blocks to the Stack, students discover why the Whole block is engineered to 84.0 mm (LCM of 1,2,3,4,6,8,12 -> 84 mm), letting 42 mm (1/2) + 21 mm (1/4) stack flush with three 21 mm (3/4) blocks.
Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fractions, Decimals & Ratios (The Pharaoh's

1. CONCRETE MANIPULATIVE SETUP

Manipulative: 84mm Fraction Studio (3x1x6U Twin Comparator Case)

Place three 1/4 blocks (3 * 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).

2. STUDENT PREDICTION (BEFORE PRESSING)

Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**0.**	Leading dot indicates 0 wholes
	**0.3**	Numerator = 3
	**0.3.**	Second dot sets fraction entry mode
	**3/4**	Denominator = 4: displays 3/4 (or 0.75)
	**3/4**	3/4 lifted to register Y
	**2**	Whole = 2
	**2.**	Dot
	**2.1**	Numerator = 1
	**2.1.**	Dot
	**2 1/2**	Denominator = 2: displays 2 1/2
	**3 1/4**	Plus combines: 3 1/4 (decimal 3.25)!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Two dots eliminate ambiguity between decimals and fractions. Physical block heights provide an infallible self-check against the 84 mm comparator pillar.

Classroom Investigation Problem:

Build 1/2 and 1/4 with blocks. Verify their combined height against three 1/4 blocks, then chec

Fraction Multiplication & The Reciprocal Key [

Fraction Multiplication & Inverses • Standard: CCSS.MATH.5.NF.B.4

1. ADVANCED INVESTIGATION SETUP

Manipulative: 84mm Fraction Studio (3x1x6U Twin Comparator Case)

Set up 3/4 of a tower (63 mm) and divide into 3 equal parts (21 mm each), taking 2 parts: (3/4) * (2/3).

2. HYPOTHESIS & MATHEMATICAL PREDICTION

3/4 * 2/3 = 6/12 = 1/2. The resulting block height matches one 1/2 block (42 mm).

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
.3.4	3/4	First fraction in Y
	3/4	Lifted
.2.3	2/3	Second fraction in X
	1/2	Multiplication gives 1/2! Exactly matches 42 mm block

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Multiplying fractions scales the height. Dividing by 2/3 is identical to multiplying by its reciprocal 3/2 using [¹/▣].

Extension Investigation Challenge:

Calculate 2/3 * 3/8 using fraction blocks and verify on the calculator. What block height match

STUDENT LAB WORKSHEET • UNIT 6

Fraction Studio: Height Equivalence & Two-Dot I

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TWO-DOT FRACTION ENTRY & TOWER EQUIVALENCE

- All fraction blocks share a 19.0 x 19.0 mm quarter-block footprint inside the 3x1x6U Twin Comparator Case. One Whole is 84.0
- 7 Divisible Fraction Families: 1 Whole (84 mm), 1/2 (42 mm), 1/3 (28 mm), 1/4 (21 mm), 1/6 (14 mm), 1/8 (10.5 mm), and 1/12
- Twin Comparator Stage: Stack equivalent fraction towers side-by-side against the 84.0 mm Whole pillar to verify equivalence

Calculator Shortcut: [. 3 . 4] = 3/4 | [2 . 1 . 2] = 2 1/2

Problem 1: Add 1/3 + 1/6 using fraction blocks and verify on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Calculate 1 1/2 - 3/4. What block height corresponds to this?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). What is the decimal equivalent of 3/8?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Multiply 2/3 * 3/4. Check against your fraction blocks.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 7 CORE LESSON / BIG IDEAS MATH (GRADES 5–6)

Chained Operations: (a + b) * (c - d) on the H

Numerical & Algebraic Expressions • Standard: CCSS.MATH.6.EE.A.2

BOOK STORY (COUNT-01): Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the 24-Tile Array Studio (where 24 packs into 1×24, 2×12, 3×8, and 4×6 rectangles), students evaluate nested expressions from the inside out.

Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Why 24 Is the Champion Packing Number) & Thinking in t

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Counting & Array Studio (COUNT-01) & 4-Line Hardware LCD Stack

Arrange Counting & Array tiles to model the two sub-expressions: (6 + 1) = 7 rows by (9 - 5) = 4 columns.

2. STUDENT PREDICTION (BEFORE PRESSING)

Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined 7 x 4 array: 7 * 4 = 28.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow:



Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**6**	Value a (6) placed into Register X
	**6**	ENTER lifts 6 into Register Y
	**1**	Value b (1) placed into Register X
	**7**	Left operator (+): Combines Y(6) + X(1) = 7 in X
	**9**	Value c (9) entered into X; 7 lifted to Y
	**9**	ENTER lifts 9 into Z and 7 into T
	**5**	Value d (5) placed into Register X
	**4**	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops t
	**28**	Root operator (*): Combines Y(7) * X(4) = 28 in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Parentheses exist in written algebra only to compensate for linear notation. On the 4-line hardware stack, precedence is intrinsic.

Classroom Investigation Problem:

Evaluate (8 + 2) * (14 - 9) by solving the inner quantities first, then checking on the hardwar

UNIT 7 EXTENSION / SINGAPORE MATH (GRADES 5-6)

Deep Stack Evaluation & Register Limits

Order of Operations & Stack Evaluation • Standard: CCSS.MATH.6.EE.A.2

1. ADVANCED INVESTIGATION SETUP

Manipulative: Counting & Array Studio (COUNT-01) & 4-Line Hardware LCD Stack

Track the 4 stack registers (T, Z, Y, X) for (12 - 4) / (2 * (1 + 1)).

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Numerator: 12 - 4 = 8. Denominator: 2 * 2 = 4. Result: 8 / 4 = 2.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
12 ENTER 4 -	8	Numerator evaluated and held in stack
2 ENTER	2	Denominator subterm pushed up to Y
1 ENTER 1 +	2	Innermost term: 1 + 1 = 2
*	4	Denominator combined: 2 * 2 = 4
/	2	8 / 4 = 2!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

The 4-register stack can evaluate nested expressions up to depth 4 without spilling to storage registers.

Extension Investigation Challenge:

Evaluate (15 - 5) / (3 + 2) by computing the numerator and denominator, then verifying on the s

STUDENT LAB WORKSHEET • UNIT 7

Postfix Evaluation & Chained Operations: The H

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: POSTFIX EVALUATION & STACK REGISTER RULES

- In postfix evaluation, operands are entered first, followed immediately by operators.
- Evaluation order matches mathematical precedence: evaluate inner sub-expressions first.
- RPN keystroke sequence evaluates complex expressions naturally without parenthesis keys.

Calculator Shortcut: Postfix Sequence: [Operand 1] [ENTER] [Operand 2] [Operator]

Problem 1: Evaluate (4 + 3) * (10 - 6) and check on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Solve (20 - 4) / (2 * 4) using inner-first evaluation.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Explain why (a + b) * c requires no parentheses in RPN.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Evaluate 5 + (3 * 4). Which operation is computed first?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 8 CORE LESSON / BIG IDEAS MATH (GRADES 7–8)

Area Modeling: Factoring x² + 5x + 6 into (x +

Polynomial Equations & Factoring • Standard: CCSS.MATH.HSA.APR.A.1

BOOK STORY (ALGB-01): From Al-Khwarizmi's Physical Squares to the 1535 Duel in Milan & Kahan's [POLY

In 820 AD in Baghdad, Al-Khwarizmi solved x² + 10x = 39 by literally arranging physical square and rectangular tiles to 'complete the square'—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! Students build physical (x + a)(x + b) tile rectangles and verify roots with the SC-32 [POLY] solver.
Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Difference of Two Squares) & Thinking in the Stack (SC

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Algebra Studio & Factoring Stage (3x1 Gridfinity, ALGB-01)

Place 1 x² tile, 5 x-tiles, and 6 unit-1 squares on the factoring stage.

2. STUDENT PREDICTION (BEFORE PRESSING)

Arrange into a solid rectangle: width is (x + 2), length is (x + 3). Total Area = x² + 5x + 6.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**1**	Coefficient a = 1 entered into X
	**1**	1 lifted to Y
	**5**	Coefficient b = 5 entered into X
	**5**	b in Y, a in Z
	**6**	Coefficient c = 6 entered into X
POLY	**-2**	Press [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The physical dimensions of the rectangle (x+2) and (x+3) correspond to roots x = -2 and x = -3 solved by the [POLY] quadratic pipeline.

Classroom Investigation Problem:

Arrange tiles for x² + 7x + 12 into a rectangle to find its factors, then verify with [POLY].

Zero Pairs: Factoring Trinomials with Negative

Factoring Quadratic Trinomials • Standard: CCSS.MATH.HSA.REI.B.4

1. ADVANCED INVESTIGATION SETUP

Manipulative: Algebra Studio & Factoring Stage (3x1 Gridfinity, ALGB-01)

Set up 1 x^2 tile and 6 negative unit tiles for $x^2 - x - 6$.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Add zero pair (+2x and -3x): forms a rectangle of (x - 3) by (x + 2).

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
1	**1**	Coefficient a = 1
ENTER	**1**	1 lifted to Y
1 +/-	**-1**	Coefficient b = -1 in X
ENTER	**-1**	b in Y, a in Z
6 +/-	**-6**	Coefficient c = -6 in X
POLY	**3**	[POLY] solves roots: Root 1 = 3 in Y (factor: x - 3), Root 2 =

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Zero-pair tiles cancel to zero while allowing rectangular dimensions to complete. [POLY] computes the real roots directly.

Extension Investigation Challenge:

Factor $x^2 - 5x + 6$ using zero pairs on the Algebra Stage and verify on the calculator.

STUDENT LAB WORKSHEET • UNIT 8

Tactile Algebra: Area Models, Polynomials & Qu

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: AREA MODELING & QUADRATIC FACTORING RULES

- Polynomial tiles represent algebraic terms: large square = x^2, rectangle = x, small square = 1.
- An algebraic expression factors when tiles form a solid rectangle with dimensions $(x + a)$ by $(x + b)$.
- Zero pairs: positive (+x) and negative (-x) tiles cancel out to zero to complete rectangular arrays.

Calculator Shortcut: [a] [ENTER] [b] [ENTER] [c] [POLY] -> Roots in Y and X registers

Problem 1: Factor $x^2 + 6x + 8$ into a rectangle of algebra tiles, then verify with [POLY].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Factor $x^2 - 4$ (difference of two squares) using zero-pair x-tiles.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Factor $x^2 + 5x + 4$ using polynomial tiles and verify.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Factor $x^2 - 5x + 6$ using zero pairs.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 9 CORE LESSON / BIG IDEAS MATH (GRADES 9–11)

Vector Magnitude & Direction: Cartesian to Pol

Right Triangle Trigonometry & Vectors • Standard: CCSS.MATH.HSN.VM.A.1

BOOK STORY (VOLV-01): A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Cro

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! With the 3-disc Unit Circle Volvelle and [▯▯,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.

Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 4: Trigonometry, Vectors & Polar Coordinates (A S

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Unit Circle Trigonometry Volvelle

Align the Unit Circle Volvelle cursor to horizontal run 4 and vertical rise 3 (forming a right triangle with hypotenuse 5).

2. STUDENT PREDICTION (BEFORE PRESSING)

Run = 4, Rise = 3. Predict magnitude: r = sqrt(4² + 3²) = 5. Predict angle: arctan(3/4) ≈ 36.87°.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
	**4**	Enter horizontal component x = 4
	**4**	4 lifted to register Y
	**3**	Enter vertical component y = 3 in X
	**36.87**	Rectangular to Polar! Screen displays angle ▯ = 36.87°
	**5.00**	Swap registers: Y displays vector magnitude r = 5.00!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The calculator's built-in polar conversion executes both the Pythagorean magnitude and arctangent angle in a single microcode step.

Classroom Investigation Problem:

A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and he

UNIT 9 EXTENSION / BIG IDEAS MATH (GRADES 9-11)

2D Vector Addition: Resultant Forces

Vector Addition & Resultant Forces • Standard: CCSS.MATH.HSN.VM.A.3

1. ADVANCED INVESTIGATION SETUP

Manipulative: Unit Circle Trigonometry Volvelle

Use the Unit Circle Volvelle to track component vectors F1 = <3, 2> and F2 = <1, 5>.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Add components: x_net = 3 + 1 = 4; y_net = 2 + 5 = 7. Magnitude r = sqrt(4² + 7²) = sqrt(65) ≈ 8.06.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
3 ENTER 1 +	4	Net horizontal force x_net = 4 in Y
2 ENTER 5 +	7	Net vertical force y_net = 7 in X
→	60.26	Resultant angle θ = 60.26°
x ⇄ y	8.06	Resultant force magnitude r = 8.06!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Vectors add component-wise. Stack registers permit component accumulation followed immediately by polar resolution.

Extension Investigation Challenge:

Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vecto

STUDENT LAB WORKSHEET • UNIT 9

Trigonometry, Vectors & Polar/Rectangular Coor

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: VECTOR DECOMPOSITION & POLAR CONVERSION RULES

- A 2D vector v = <x, y> has horizontal run x in Y and vertical rise y in X.
- Rectangular to Polar [yellow] [▯▯_r]: Converts (x, y) into angle ▯ (in X) and magnitude r (in Y).
- Trig functions [sin], [cos], [tan] operate on the angle in register X.

Calculator Shortcut: [RUN x] [ENTER] [RISE y] [yellow] [▯▯_r] -> Angle in X, Magnitude in Y

Problem 1: Find magnitude and angle for vector <12, 5>.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Find the hypotenuse of a right triangle with legs 7 and 24.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Calculate sin(30°) on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 10 CORE LESSON / BIG IDEAS MATH (GRADES 10–12)

Continuous Compounding & Euler's Constant e

Exponential & Logarithmic Functions • Standard: CCSS.MATH.HSF.LE.A.4

BOOK STORY (PRIME-01): The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Eule

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 * log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.
Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 3: Transcendentals (John Napier & Euler) & Ch. 11

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Fraction Studio (Proportional Doubling Towers)

Stack Fraction Blocks representing exponential doubling: 12 mm, 24 mm, 48 mm, 96 mm.

2. STUDENT PREDICTION (BEFORE PRESSING)

Invest $1,000 at 7% continuous interest for 10 years: A = 1000 * e^(0.07 * 10) = 1000 * e^0.7. Will it double? (e^0.7 ≈ 2.01).

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
0.07	**0.07**	Rate r = 0.07
STO	**0.07**	Rate in Y
10	**10**	Time t = 10 in X
.	**0.7**	Exponent r * t = 0.70
e^x	**2.0138**	e^0.70 = 2.01375 (Growth factor: more than doubled!)
1000	**1000**	Principal P = 1000 (growth factor lifts to Y)
.	**2013.75**	Total accumulated amount A = $2,013.75!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack computes nested transcedentals from innermost exponent out to outer multiplier without parenthesis keys.

Classroom Investigation Problem:

A bacteria culture starts at 500 and triples every hour. Use [y□] to calculate population after

UNIT 10 EXTENSION / BIG IDEAS MATH (GRADES 10–12)

Solving for Time: Doubling Time & Natural Log

Logarithms & Inverting Exponentials • Standard: CCSS.MATH.HSF.BF.B.5

1. ADVANCED INVESTIGATION SETUP

Manipulative: Fraction Studio (Proportional Doubling Towers)

Configure stack registers on the 4-line LCD: target multiplier in Y, growth rate in X.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

At 5% growth (r = 0.05), doubling time is t = ln(2) / 0.05 ≈ 0.69315 / 0.05 ≈ 13.86 years.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
2	**2**	Doubling target factor 2
LN	**0.6931**	Natural log ln(2) = 0.693147... in X
0.05	**0.05**	Rate r = 0.05 (ln(2) lifts to Y)
÷	**13.86**	t = ln(2) / r = 13.86 years!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

The natural log [LN] is the exact inverse of [e☒]. RPN preserves full floating point precision during the division.

Extension Investigation Challenge:

A radioactive isotope has decay constant k = 0.035. Find its half-life: t = ln(2) / k.

STUDENT LAB WORKSHEET • UNIT 10

Exponential Growth, Compound Interest & Logari

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Exponential growth formula: A = P * e^(r*t) for continuous growth, or A = P * (1 + r/n)^(n*t).
- Natural exponential [e□] raises Euler's constant e to power X.
- Natural logarithm [LN] inverts exponential growth: ln(e□) = x.

Calculator Shortcut: [r] [ENTER] [t] [*] [e□] [P] [*] = [ACCUMULATED AMOUNT A]

Problem 1: Calculate e^2 on the calculator to 4 decimal places.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: Calculate doubling time for an investment growing at 8% continuous interest.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Compute 10^3.5 using [yellow] [10□].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Evaluate log10(500) using [yellow] [LOG].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 11 CORE LESSON / AP STATISTICS (GRADES 11–12 / AP)

Combinations vs. Permutations: Subsets & Order

Probability, Random Variables & Combinatorics • Standard: CCSS.MATH.HSS.CP.B.9

BOOK STORY (PROB-01): Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brew

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students roll fair 16 mm polyhedral dice and accumulate trials with [⊠+].

Companion Book: From Blocks to the Stack (SC-6 Ch. 7: Galton's Bean Machine) & Thinking in the Stack (SC-32

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Probability Dice & Modular Spinner Set

Use the 4 precision dice and 10-sector spinner to model multi-stage discrete outcomes.

2. STUDENT PREDICTION (BEFORE PRESSING)

How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! * 5!) = 40320 / (6 * 120) = 56.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
8	**8**	Total candidates n = 8
x!	**40320**	8! = 40,320 in register X
3	**3**	Subset r = 3 (40320 lifts to Y)
x!	**6**	3! = 6 in X
·	**6720**	40320 / 6 = 6720 in X
5	**5**	Remaining n - r = 5 (6720 lifts to Y)
x!	**120**	5! = 120 in X
·	**56**	6720 / 120 = 56 combinations!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack handles multi-step factorial simplification effortlessly, preventing arithmetic overflow.

Classroom Investigation Problem:

A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

UNIT 11 EXTENSION / AP STATISTICS (GRADES 11–12 / AP)

Statistical Accumulation: Mean & Variance with

One-Variable Data Analysis & Summations • Standard: CCSS.MATH.HSS.ID.A.2

1. ADVANCED INVESTIGATION SETUP

Manipulative: Probability Dice & Modular Spinner Set (Statistical Data Generation)

Roll precision dice to generate discrete data points for [Σ+] accumulation: 4, 6, 7, 8, 10.

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean $\bar{x}$ = 35 / 5 = 7.0.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
4 sigma_plus	1	First score entered: sample size n = 1
6 sigma_plus	2	Second score: n = 2
7 sigma_plus	3	Third score: n = 3
8 sigma_plus	4	Fourth score: n = 4
10 sigma_plus	5	Fifth score: n = 5 total data points

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

The [Σ+] key accumulates sum(x), sum(x^2), and count n into dedicated background statistical registers.

Extension Investigation Challenge:

Accumulate data points 12, 15, 18, 21 using [Σ+]. What is the sample count n and mean?

STUDENT LAB WORKSHEET • UNIT 11

Combinatorics, Probability & Series Accumulati

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: COMBINATIONS, PERMUTATIONS & STACK FACTORIALS

- Factorial [yellow] [x!] multiplies n * (n-1) * ... * 1.
- Permutations (order matters): nPr = n! / (n - r)!.
- Combinations (subsets/groups): nCr = n! / (r! * (n - r)!).

Calculator Shortcut: [n] [yellow] [x!] [r] [yellow] [x!] [÷] [n - r] [yellow] [x!] [÷] = nCr

Problem 1: Calculate 6! (6 factorial) on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 3: Calculate 7C2 (combinations of 2 from 7).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

Problem 4: Accumulate 5 and 15 with [⊠+]. What is the mean?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction:

Final Answer:

[ ] Checked on StackCalc

UNIT 12 CORE LESSON / AP COMPUTER SCIENCE PRINCIPLES (GRADES 9–12 / AP CS)

Number Representation: Decimal to Binary and H

Data Representation & Binary Number Systems • Standard: CCSS.MATH.PRACTICE.MP5

BOOK STORY (BASE-10): Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-B

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using an 8-slot Ten-Frame register and [BASE], students decode bytes, hex nibbles, and bitwise masks.

Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 10: Computer Science, Base-N & 36-Bit Word Logic

1. CONCRETE MANIPULATIVE SETUP

Manipulative: Ten-Frame Stage (4x2 Gridfinity)

Use the 8 wells of the Ten-Frame as an 8-bit byte. Place pegs in wells for active '1' bits, leave empty for '0'.

2. STUDENT PREDICTION (BEFORE PRESSING)

Place pegs at positions 0, 1, and 3: binary 00001011_2 = 8 + 2 + 1 = 11_10. Predict hex value: 0B_16.

3. CALCULATOR ACTION (HARDWARE TRAPEZOID KEYCAPS)

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
11	**11**	Enter decimal integer 11 in DEC mode
	HEX 0B	Press [yellow] [BASE]: screen displays Hexadecimal 0B!
	BIN 1011	Press [yellow] [BASE]: screen displays Binary 1011!
	OCT 13	Press [yellow] [BASE]: screen displays Octal 13!
	DEC 11	Press [yellow] [BASE]: returns to Decimal 11

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Computer memory stores identical bit patterns; base conversions alter only the human visual representation.

Classroom Investigation Problem:

Enter decimal 255. Use [yellow] [BASE] to convert to HEX and BIN. What is the binary pattern?

UNIT 12 EXTENSION / AP COMPUTER SCIENCE PRINCIPLES (GRADES 9-12 / AP CS)

Bitwise Logic: AND Masking & Network Subnets

Boolean Logic & Bitwise Operations • Standard: CCSS.MATH.PRACTICE.MP5

1. ADVANCED INVESTIGATION SETUP

Manipulative: Ten-Frame Stage (4x2 Gridfinity)

Row 1 holds byte A (1100 = 12). Row 2 holds mask B (1010 = 10).

2. HYPOTHESIS & MATHEMATICAL PREDICTION

Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

3. CALCULATOR ACTION & REGISTER TRANSFORMATION

Keystroke Flow: 

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
12	12	Enter operand A = 12 (binary 1100)
	12	12 lifted to register Y
10	10	Enter operand B = 10 (binary 1010) in X
and	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

4. EXTENSION ANALYSIS & DEEPER CONNECTIONS

Bitwise AND masks bits, clearing unwanted bits to zero. This is the exact operation used in IPv4 subnet masking.

Extension Investigation Challenge:

Calculate bitwise OR for 12 (1100) and 3 (0011). Predict the binary and decimal result.

STUDENT LAB WORKSHEET • UNIT 12

Digital Logic, Base Conversions & Computer Arc

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: NUMBER BASES & BITWISE LOGIC GATES

- Binary (Base 2): digits 0 and 1. Hexadecimal (Base 16): digits 0-9 and A-F.
- Press [yellow] [BASE] to cycle between DEC (decimal), HEX (hexadecimal), OCT (octal), and BIN (binary).
- Bitwise AND: result bit is 1 only if BOTH bits are 1 (used for masking and subnets).

Calculator Shortcut: [DECIMAL VALUE] [yellow] [BASE] -> Cycles to HEX, BIN, OCT

Problem 1: Convert decimal 42 to binary using [yellow] [BASE].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 2: Convert hexadecimal 2A to decimal.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

Problem 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on StackCalc

TEACHER SOLUTION GUIDE

Master Solutions & Keystrokes — Part 1

Complete worked solutions, student practice answers, and verification keystrokes.

Unit 1: Counting, Cardinality & The One-More Step Counter (Pre-K – Kindergarten • SC-6 Elementary (Grades K-6))

Core Lesson: 5 pegs. Keystrokes: [4] [ENTER] [1] [+]

Core Check: 

Extension: 4 more pegs. Keystrokes: [6] [ENTER] [2] [-]

Ext. Check: 

Lab Sheet Key: P1: 5 | P2: 3 | P3: 0 (Zero) | P4: 4

Unit 2: Part-Part-Whole & Doubles: The Math Domino Studio (Grades K-1 • SC-6 Elementary (Grades K-6))

Core Lesson: 9 pips. Keystrokes: [5] [ENTER] [4] [+]

Core Check: 

Extension: Missing Part is 3. Keystrokes: [8] [ENTER] [5] [-]

Ext. Check: 

Lab Sheet Key: P1: 4 | P2: 9 | P3: 9 | P4: 3

Unit 3: Ten-Frames: Complements to 10 & 5-Structure (Grade 1 • SC-6 Elementary (Grades K-6))

Core Lesson: 4 counters. Keystrokes: [1][0] [ENTER] [6] [-]

Core Check: 

Extension: 15. Keystrokes: [9] [ENTER] [1] [+] [5] [+]

Ext. Check: 

Lab Sheet Key: P1: 6 | P2: 12 | P3: 1 | P4: 14

Unit 4: Number Racks (Rekenrek): 5-Structure & Doubles (Grades 1-2 • SC-6 Elementary (Grades K-6))

Core Lesson: 14 beads. Keystrokes: [7] [ENTER] [+]

Core Check: 

Extension: 13. Keystrokes: [6] [ENTER] [+] [1] [+]

Ext. Check: 

Lab Sheet Key: P1: 16 | P2: 17 | P3: 18 | P4: 11

Unit 5: The 4-Register Stack: Four Operations & Automatic Lift (Grades 3-4 • SC-6 Elementary (Grades K-6))

Core Lesson: 3 cookies per bag. Keystrokes: [1][5] [ENTER] [6] [-] [3] [÷]

Core Check: 

Extension: 19. Keystrokes: [2][4] [ENTER] [6] [÷] [1][5] [+]

Ext. Check: 

Lab Sheet Key: P1: 27 | P2: 6 | P3: 4 | P4: 3 eggs

Unit 6: Fraction Studio: Height Equivalence & Two-Dot Notation (Grades 4-5 • SC-6 Elementary (Grades K-6))

Core Lesson: 3/4 (height 63 mm). Keystrokes: [.][1][.][2] [ENTER] [.][1][.][4] [+]

Core Check: 

Extension: 1/4 (height 21 mm). Keystrokes: [.][2][.][3] [ENTER] [.][3][.][8] [×]

Ext. Check: 

Lab Sheet Key: P1: 1/2 (42 mm) | P2: 3/4 (63 mm) | P3: 0.375 | P4: 1/2 (42 mm)

Scoring Criteria

Award credit for: (1) Correct concrete manipulative modeling, (2) Written mathematical strategy & prediction prior to calculator check, and (3) Verification on the StackCalc hardware or simulator.

TEACHER SOLUTION GUIDE

Master Solutions & Keystrokes — Part 2

Complete worked solutions, student practice answers, and verification keystrokes.

Unit 7: Postfix Evaluation & Chained Operations: The Hardware Stack (Grades 5-6 • SC-6 Elementary (Grades K-6))

Core Lesson: 50. Keystrokes: [8] [ENTER] [2] [+] [1][4] [ENTER] [9] [-] [×]

Core Check: 

Extension: 2. Keystrokes: [1][5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Ext. Check: 

Lab Sheet Key: P1: 28 | P2: 2 | P3: a b + c * | P4: 17

Unit 8: Tactile Algebra: Area Models, Polynomials & Quadratic Factoring (Grades 7-8 • SC-6 Elementary & SC-32 Scientific B

Core Lesson: (x + 3)(x + 4), roots -3 and -4. Keystrokes: [1] [ENTER] [7] [ENTER] [1][2] [POLY]

Core Check: 

Extension: (x - 2)(x - 3), roots 2 and 3. Keystrokes: [1] [ENTER] [5] [+/-] [ENTER] [6] [POLY]

Ext. Check: 

Lab Sheet Key: P1: (x + 2)(x + 4); roots -2, -4 | P2: (x - 2)(x + 2); roots 2, -2 | P3: (x + 1)(x + 4); roo

Unit 9: Trigonometry, Vectors & Polar/Rectangular Coordinates (Grades 9-11 • SC-32 Scientific (Grades 7-12+))

Core Lesson: Displacement: 100 m, Heading: 53.13°. Keystrokes: [6][0] [ENTER] [8][0] [yellow] [▯▯_r] [x▯y]

Core Check: 

Extension: Magnitude: 6.71. Resultant: <3, 6>.

Ext. Check: 

Lab Sheet Key: P1: ▯ = 22.62°, r = 13 | P2: 25 | P3: 0.5 | P4: x = 7.07, y = 7.07

Unit 10: Exponential Growth, Compound Interest & Logarithms (Grades 10-12 • SC-32 Scientific (Grades 7-12+))

Core Lesson: 40,500 bacteria. Keystrokes: [3] [ENTER] [4] [y▯] [5][0][0] [×]

Core Check: 

Extension: 19.80 years. Keystrokes: [2] [LN] [0][.][0][3][5] [÷]

Ext. Check: 

Lab Sheet Key: P1: 7.3891 | P2: 8.66 years | P3: 3162.28 | P4: 2.6990

Unit 11: Combinatorics, Probability & Series Accumulation (Grades 11-12 / AP • SC-32 Scientific (Grades 7-12+))

Core Lesson: 210 combinations. Keystrokes: [1][0] [x!] [4] [x!] [÷] [6] [x!] [÷]

Core Check: 

Extension: Count n = 4, Mean $\bar{x}$ = 16.5.

Ext. Check: 

Lab Sheet Key: P1: 720 | P2: 60 | P3: 21 | P4: Mean = 10

Unit 12: Digital Logic, Base Conversions & Computer Architecture (Grades 9-12 / AP CS • SC-32 Scientific (Grades 7-12+))

Core Lesson: HEX: FF, BIN: 11111111 (8 ones). Keystrokes: [2][5][5] [yellow] [BASE]

Core Check: 

Extension: Binary: 1111, Decimal: 15. Keystrokes: [1][2] [ENTER] [3] [OR]

Ext. Check: 

Lab Sheet Key: P1: 101010 (32 + 8 + 2) | P2: 42 | P3: 7 (0111) | P4: 6 (0110)

Scoring Criteria

Award credit for: (1) Correct concrete manipulative modeling, (2) Written mathematical strategy & prediction prior to calculator check, and (3) Verification on the StackCalc hardware or simulator.

Fabrication & Slicer Profiles

Guidelines for 3D printing the modular Gridfinity Learning Lab set in PLA or PETG.

Material Selection

PLA or PETG. Standard PLA provides crisp debossed text definition. PETG offers superior impact resistance for frequent classroom handling. No multi-material system or laser required.

Slicer Profile Recommendations

Layer height: 0.20 mm. Perimeters: 3. Infill: 15-20% gyroid. Top/bottom solid layers: 4. The 0.6 mm inset debossing creates exactly 3 solid recessed layers for high visual contrast under classroom lighting.

Gridfinity Tolerances

Trays use Zack Freedman's 42.0 mm pitch with standard 0.5 mm bin clearances and 4.0 mm base profiles. Always print the 'fit_coupon.stl' first to test socket clearances on your printer.

Storage & Organization

The 4x4 baseplate (168 x 168 mm) acts as both the activity dock and the storage foundation. Trays stack cleanly into classroom cubbies, completely eliminating disposable cardboard packaging.

Hardware Keycap Parity

All keycaps in this guide are rendered directly from StackCalc firmware simulator definitions (Firmware/Simulator/labels.json) ensuring 100% exact parity with the physical and watch calculator.

WHERE TO GET SPIRAL BOOKS, FLAGSHIP BOOKS & MANIPULATIVE BLOCKS (5 CHANNELS)

- **Spiral-Bound Curriculum Books & Workbooks (Lulu POD): https://www.lulu.com/spotlight/stackcalc**
- Flagship Trade Books — Paperback & Hardcover (Amazon): https://link.amazon/B0f5cP0Ur
- Handcrafted 3D-Printed Manipulative Boxed Kits & Hardware (Etsy): https://stackcalcmath.etsy.com
- On-Demand Factory Printed Manipulative Kits & Refills (Slant 3D): https://teleportpod.com/portals/profile/StackCalc
- 100% Free Open-Source STLs & 3MF Print Plates (Printables): https://www.printables.com/@StackCalc_5593834

Manipulative Kit BOM & Millimeter Specs

Complete classroom inventory, Gridfinity footprints, piece counts, and physical-to-stack invariants.

SKU	Manipulative Studio Name	Gridfinity Footprint & Dimensions	Core Physical Invariant
BASE-10	Ten-Frame & Two-Tone Studio	4x2 Base + Two 2x5 Trays (84×168 mm)	5+5 subitizing & 2-color zero-pair c
DOMI-01	Math Domino Part-Whole Studio	3x2 Tray + 28 Tactile Dominoes	Part-part-whole number bonds & doubl
RACK-01	Number Rack (Rekenrek) Stage	4x2 Dual-Row 5+5 Sliding Stage	Two-row 5+5 structure mirrors Y & X
PLACE-01	Place Value Base-10 Studio	3x2 H-T-O Stage + 100/10/1 Blocks	10:1 decimal trading & 4-register li
COUNT-01	Counting & Array Studio	3x2 Array Frame + Interlocking Tiles	m×n rectangular area & factor pairs
FRAC-01	84mm Fraction Comparator	3x1x6U Twin Channel + 39 Towers	1 Whole = 84.0 mm (divisible by 2,3,
PRIME-01	Prime Factor Log Studio	3x2 Log Stage + Prime Towers	Height = k·ln(p); stacking adds logs
TIME-01	Modulo-60 Geared Clock	3x3 Epicyclic 12:1 Planetary Clock	True 12:1 mechanical gear ratio & mo
ALGB-01	Tactile Algebra Studio	3x1x6U Case + x^2, x & 1 Area Tiles	2D area rectangle side lengths = fac
PROB-01	Probability Dice & Spinner	3x3 Rolling Tray + Fair Dice & Dial	Isohedral symmetry & empirical bell
VOLV-01	Unit Circle Trig Volvelle	120 mm 3-Disc Rotary Instrument	Aperture windows reveal exact (cos □
QUEST-01	SC-6 Quest Scoreboard Kit	3x1x7U Token Box + 6×2 Snap Lid	60 universal [+10] tokens; 2 stacked

Zero-Inventory Replacement Parts Policy

Lost a fraction block or token in class? Download the individual STL 100% free on Printables (@StackCalc_5593834) or order a single factory-printed replacement part on Slant 3D.

BACK MATTER • APPENDIX C: CALCULATOR QUICK-REFERENCE

SC-6 & SC-32 Master Keystroke Reference

The 4-register RPN stack (T, Z, Y, X) and essential elementary, middle, and high school verification sequences.

THE 4-REGISTER RPN STACK ELEVATOR (T, Z, Y, X) — HOW AUTOMATIC STACK LIFT WORKS

- Register X (Floor 1 — Active Entry): Holds the number currently being typed or the result of the last operation.
- Register Y (Floor 2 — Waiting Operand): Holds the first number after you press [ENTER]. Binary keys (+, −, ×, ÷) combine Y and X.
- Registers Z & T (Floors 3 & 4): Hold intermediate results during chained multi-step expressions—zero parentheses ever needed!
- **Automatic Stack Lift: After any calculation (+, −, ×, ÷, x^2, $\sqrt{x}$), typing a new number automatically lifts the old result to Y!**

Mathematical Operation	Instrument	Canonical Keystroke Sequence	Display / Stack Result
Basic Addition (8 + 5)	SC-6 / SC-32	[8] [ENTER] [5] [+]	X = 13 (T:0, Z:0, Y:0)
Chained Expression (3×4)+(5×6)	SC-6 / SC-32	[3] [ENTER] [4] [×] [5] [ENTER] [6]	X = 42 (Y holds 12 while 5×6 run
Skip-Counting Fact Ladder (+3s)	SC-6	[3] [ENTER] [3] [g][REP] [+] [+] [+]	X steps 6 -> 9 -> 12 -> 15
Quotient & Remainder (23 ÷ 4)	SC-6	[23] [ENTER] [4] [g][÷R]	Y = 5 (Quotient), X = 3 (Rem)
Two-Dot Proper Fraction (2/3 + 1	SC-6 / SC-32	[.] [2] [.] [3] [ENTER] [.] [1] [.]	X = 11/12 (Exact rational)
Mixed Fraction Entry (2 and 3/8)	SC-6 / SC-32	[2] [.] [3] [.] [8] [ENTER]	X = 2 3/8 (or 19/8 via [f][d/c])
Prime Factorization & GCF / LCM	SC-6	[24] [g][FACT] | [24] [ENTER] [36]	$2^3 \times 3^1$ | GCF = 12, [f][LCM] = 7
Equation Balance Check (4×9+5 =	SC-6	[41] [ENTER] [4] [ENTER] [9] [×] [5]	Flashes NO (proving 41 = 41!)
Mean, Median, IQR & MAD	SC-6 / SC-32	[g][CL⊠] x1 [⊠+] x2 [⊠+] ... [g][x]	Exact Mean, Median, [IQR] & [g][
2D Vector Polar Conversion (3, 4	SC-32	[4] [ENTER] [3] [⊠][⊠⊠,r]	X = 5.0000 (r), Y = 53.1301° (⊠)
Quadratic / Cubic Polynomial Roo	SC-32	[a] [ENTER] [b] [ENTER] [c] [⊠][POLY	Real or complex roots in X, Y, Z
Base-N Binary / Hex Conversion	SC-32	[42] [⊠][BASE] [BIN] / [HEX]	00101010_2 / 0x2A (36-bit word)

Classroom Pacing & Station Tracker

Photocopiable yearly planning log and station observation notes for teachers and homeschool parents.

Unit & Topic	Kit SKU	Date Taught	Core / Ext / Lab	Class Mastery Notes
Unit 01: Counting, Cardinality & The On	BASE-10	____/____/20__	[ ] C [ ] E [ ] L	
Unit 02: Part-Part-Whole & Doubles: The	DOMI-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 03: Ten-Frames: Complements to 10	BASE-10	____/____/20__	[ ] C [ ] E [ ] L	
Unit 04: Number Racks (Rekenrek): 5-Str	RACK-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 05: The 4-Register Stack: Four Ope	PLACE-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 06: Fraction Studio: Height Equiva	FRAC-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 07: Postfix Evaluation & Chained O	COUNT-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 08: Tactile Algebra: Area Models,	ALGB-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 09: Trigonometry, Vectors & Polar/	VOLV-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 10: Exponential Growth, Compound I	PRIME-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 11: Combinatorics, Probability & S	PROB-01	____/____/20__	[ ] C [ ] E [ ] L	
Unit 12: Digital Logic, Base Conversion	BASE-10	____/____/20__	[ ] C [ ] E [ ] L	

TEACHER OBSERVATION NOTES, MISCONCEPTION LOG & 3D PRINT REFILL CHECKLIST